90+7′: The Economics of FIFA's Stoppage-Time Directive*

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Abstract

In November 2022, FIFA instructed match officials to fully compensate for time lost during professional football matches—enforcing a rule systematically under-applied for decades. We study this enforcement shift across 15 European leagues and 36,000 matches. The directive triggered a near-universal norm shift: enforcement intensity tracked institutional proximity to FIFA, producing a dose-response gradient from the Premier League ($d = 1.13$) to Serie B ($d \approx 0$). The most striking consequence is goal *displacement*: total goals per match were unchanged ($\beta = -0.033$, $p = 0.39$), but goals migrated from the final minutes of regulation into expanded stoppage time, raising the share of matches with a result-changing stoppage-time goal from 7.3% to 10.3%. A simple model of optimal attacking intensity under known remaining time rationalises this pattern. Compliance was near-universal (95% of referees individually increased stoppage time), achieved through focal-point coordination rather than monitoring or sanctions.

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1 Introduction

Professional football matches nominally last 90 minutes, but the actual duration has always been at the referee's discretion. Law 7 of the Laws of the Game has required referees to add time lost to interruptions since 1886, yet for decades the rule was systematically under-enforced: added time averaged three to five minutes regardless of actual dead time, which typically exceeded 28 minutes per match (Forcher et al., 2024). In November 2022, Pierluigi Collina, chairman of FIFA's Referees Committee, instructed World Cup officials to restore the actual time lost. The resulting matches—exceeding 100 minutes before a global audience of five billion—created a new public benchmark for proper enforcement.

This paper studies what happened next, using data from 15 European leagues and over 36,000 matches. We document three findings that speak to broader questions in economics.

First, the directive produced a dose-response gradient in which enforcement intensity tracked institutional proximity to FIFA with remarkable precision. The five major European leagues (the "Big Five": England's Premier League, Spain's La Liga, Germany's Bundesliga, Italy's Serie A, and France's Ligue 1) were formally addressed, but the signal propagated far beyond them. The Premier League shifted by 2.1 minutes (Cohen's $d = 1.13$); the Scottish Premiership—nominally untreated—shifted by 1.5 minutes ($d = 0.83$); Serie A, despite formal treatment, shifted by only 0.7 minutes ($d = 0.40$). This gradient reveals how a global governing body's signal travels through the decentralised architecture of European football, attenuating at each institutional boundary.

Second, and most distinctively, the additional minutes produced goal *displacement* rather than goal creation. Total goals per match were unchanged ($\hat{\beta} = -0.033$, $p = 0.39$), a precisely estimated null. But goals migrated from the final minutes of regulation into the expanded stoppage window: per-minute scoring rates in the 76–84 minute window fell by 8.6%, while the 90+ window rose by 35.5%. The share of matches in which the result changed during stoppage time rose from 7.3% to 10.3%—roughly 55 additional decisive moments per Big Five season. We develop a simple model of optimal attacking intensity under known remaining

time to rationalise this finding: when the trailing team expects more time, it substitutes patience for desperation, spreading attacking effort across a longer horizon without increasing aggregate intensity.

Third, compliance was near-universal and immediate. Among referees with sufficient pre- and post-directive observations, 95–100% individually increased their stoppage time, with the coefficient of variation across referees declining from 0.17 to 0.14. The shift was permanent: three full seasons later, there has been no reversion. The mechanism was focal-point coordination (Schelling, 1960), not monitoring or sanctions—the World Cup demonstration crystallised a new behavioural norm that became self-reinforcing once a critical mass adopted it.

These findings contribute to several literatures. In sports economics, the closest antecedents study referee bias under social pressure (Garicano et al., 2005; Dohmen, 2008; Price and Wolfers, 2010) and strategic manipulation of contests (Duggan and Levitt, 2002). Our setting differs in that the rule was unchanged—only its enforcement shifted—allowing us to study compliance with an existing but under-enforced regulation, a problem common across regulatory domains (Duflo et al., 2013; Johnson, 2020). The displacement result connects to the labour economics literature on working hours and productivity (Pencavel, 2015): adding minutes to a football match is analogous to extending a workday, and the finding of constant returns to time has implications for how output responds to time extensions when effort is endogenous. The compliance analysis speaks to the literature on norm evolution and focal-point coordination (Young, 1993; Roth and Ockenfels, 2002), showing how a single high-visibility demonstration can resolve a coordination failure among hundreds of independent professionals.

The paper proceeds as follows. Section 2 documents the enforcement gradient across 15 leagues. Section 3 presents our central finding—goal displacement—and develops a model of optimal attacking intensity that rationalises it. Section 4 describes the data. Section 5 presents the identification strategy, including a dose-response framework that addresses the

contamination inherent in the binary difference-in-differences. Section 6 documents referee compliance and convergence. Section 7 concludes.

2 The First Stage: What Happened to Match Time

Law 7 of the Laws of the Game has required referees to add time lost to interruptions since 1886, yet for decades the rule was systematically under-enforced: referees added three to five minutes regardless of actual dead time, which typically exceeded 28 minutes per match. In November 2022, Pierluigi Collina, chairman of FIFA's Referees Committee, instructed World Cup officials to restore the *actual* time lost—producing matches exceeding 100 minutes before a global audience of five billion viewers. IFAB endorsed the practice at its March 2023 Annual General Meeting, using "should" rather than "must," preserving each federation's discretion and creating the institutional heterogeneity that drives our empirical design.

After the World Cup, matches across Europe got longer—but not uniformly, and not everywhere. The enforcement gradient that emerged tracks institutional proximity to FIFA's refereeing apparatus with remarkable fidelity, revealing how a global governing body's signal travels through the decentralised architecture of European football.

We study 15 European leagues spanning the five major top-flight competitions (the "Big Five": England's Premier League, Spain's La Liga, Germany's Bundesliga, Italy's Serie A, and France's Ligue 1), their associated second divisions where data are available, and several smaller top-flight leagues.

2.1 What Changed, and How Much

The headline is unambiguous. Across the five treated leagues, mean second-half stoppage time rose from 4.99 minutes (N = 10,763) to 6.56 minutes (N = 4,596), an increase of 1.57 minutes. The entire distribution shifted rightward: a Kolmogorov-Smirnov test rejects the null that the pre- and post-directive samples share a common distribution ($D = 0.332$,

Table 1: Enforcement Intensity by League

League	Code	Pre Mean	Post Mean	Δ	Cohen's d	Intensity
Premier League†	E0	5.39	7.50	+2.11	1.13	High
Ligue 1†	F1	4.71	6.50	+1.79	1.05	High
Bundesliga†	D1	4.34	6.12	+1.78	0.92	High
Scottish Premiership	SC0	2.83	4.30	+1.47	0.83	High
La Liga†	SP1	5.18	6.64	+1.46	0.70	High
Belgian Pro League	B1	3.09	4.40	+1.31	0.66	High
2. Bundesliga	D2	3.95	4.98	+1.04	0.59	High
Championship	E1	5.67	6.70	+1.03	0.59	High
Primeira Liga	P1	5.97	6.94	+0.97	0.42	Moderate
Ligue 2	F2	2.81	3.76	+0.95	0.56	Moderate
Greek Super League	G1	3.58	4.41	+0.82	0.38	Moderate
Serie A†	I1	5.18	5.91	+0.73	0.40	Moderate
Eredivisie	N1	5.24	5.71	+0.47	0.23	Low
Serie B	I2	4.05	3.97	-0.08	-0.04	Low

Notes: Pre = seasons before 2023-24. Post = 2023-24 onward. Δ = Post − Pre mean second-half stoppage time (minutes). Cohen's d = standardized effect size. Intensity classified as High ($\Delta > 1.0$), Moderate ($0.5 < \Delta \leq 1.0$), Low ($\Delta \leq 0.5$). † denotes Big Five league.

$p < 10^{-10}$). The probability of exceeding seven minutes of added time rose from 14.8% to 39.2%; the probability of exceeding ten minutes—once a rarity that generated tabloid headlines—rose from 1.8% to 7.8% (Figure 2).

The standard deviation widened from 1.79 to 2.24 minutes. The new norm did not replace one rigid convention with another. The fourth official's added-time board, which once displayed a narrow range, now shows wider variation, reflecting finer differentiation among match circumstances. The directive asked referees to *measure* rather than *estimate*, and measurement produces more variance than convention.

Table 1 reports the league-by-league breakdown, ordered by Cohen's d.

The gradient is steep: the Premier League shifted by 2.11 minutes ($d = 1.13$), more than a full standard deviation above the pre-directive mean; Serie B showed no measurable change ($\Delta = -0.08$, $d = -0.04$). Between these poles, thirteen leagues arranged themselves along a continuum that tracks institutional proximity to FIFA with remarkable precision (Figures 1 and 3).

2.2 How the Signal Traveled

The signal originated with Pierluigi Collina. As chairman of FIFA's Referees Committee, Collina is simultaneously the sport's most recognizable official and its most powerful referee administrator. When he told the 36 match officials assembled for Qatar that every lost second must be restored, the instruction carried the weight of both formal authority and unmatched personal credibility. England–Iran ran to 117 minutes. The World Cup's cumulative viewership exceeded five billion, and every viewer saw what the new standard looked like: fourth officials holding up boards reading "+10," "+12," "+14."

Those referees returned to domestic duty in January 2023. Approximately 35 were members of UEFA's Elite Panel; their visible adoption of the new norm in league matches broadcast to millions created a public expectation that the rest of each country's referee corps would follow. The formal institutional step came at IFAB's Annual General Meeting in March 2023, where the board endorsed the practice—using "should," not "must," and thereby preserving each federation's discretion over implementation.

From IFAB, the signal passed through three visible channels. *Direct referee exposure*: officials who worked the World Cup or regularly officiate FIFA fixtures received the instruction firsthand and carried it home. *Federation guidance*: pre-season briefings, updated assessor protocols, and formal memoranda translated IFAB's recommendation into operational standards. The assessor's report—the post-match evaluation that grades referee performance and influences future assignments—is the critical lever; when assessors began flagging insufficient stoppage time as a deficiency, the incentive facing every referee in the pool shifted overnight. *Media pressure*: in leagues with large broadcast audiences, any referee who conspicuously failed to add adequate time faced immediate public criticism. The pundit's question—"Why only four minutes when there were three VAR checks?"—became a form of decentralized enforcement requiring no monitoring apparatus and no sanctions.

2.3 Within-Country Pairs

The most revealing comparisons are within countries, where top-flight and second-division leagues share a national federation but differ in their connection to FIFA's referee infrastructure.

England: PGMOL versus the EFL. The Premier League added 2.11 minutes ($d = 1.13$); the Championship added 1.03 minutes ($d = 0.59$). The gap of more than a full minute per match reflects the institutional divide between two referee organizations operating under the same Football Association. The Professional Game Match Officials Limited (PGMOL) manages Premier League referees as full-time professionals—a status they have held since 2001, when the FA abandoned the pretense that officials overseeing matches worth hundreds of millions of pounds could do so as a weekend avocation. PGMOL referees receive centralized briefings, structured assessor reviews, and regular international assignments that connect them directly to FIFA. The most experienced were at the World Cup or serve on UEFA's Elite Panel; they returned to domestic duty already calibrated to the new standard. Championship referees, managed by the EFL, sit one rung below: not full-time professionals, but absorbing PGMOL norms through the development pipeline, as the most promising are groomed for promotion to the top-flight list. The Championship's substantial shift confirms the norm penetrated beyond the treated tier; the 1.08-minute gap confirms the signal attenuated at the organizational boundary between the two bodies.

Germany: the DFB pipeline. The Bundesliga shifted by 1.78 minutes ($d = 0.92$); the 2. Bundesliga by 1.04 minutes ($d = 0.59$). Germany's referee governance is more centralized than England's: the Deutscher Fußball-Bund (DFB) oversees both competitions through a unified referee department, and promising second-division referees are developed explicitly for Bundesliga promotion through a structured pathway. The within-Germany gap (0.74 minutes) is smaller than the within-England gap (1.08 minutes), consistent with

the DFB's unified pipeline transmitting the enforcement signal more efficiently to its second division than the FA's bifurcated structure manages.

France: shared officials. Ligue 1 added 1.79 minutes ($d = 1.05$); Ligue 2 added 0.95 minutes ($d = 0.56$). French refereeing presents a distinctive institutional feature: the Direction Technique de l'Arbitrage (DTA) manages officiating for both divisions, and some referees work matches in both competitions within a single season. This partial overlap in referee pools means the enforcement signal did not need to cross an organizational boundary to reach Ligue 2—it arrived through the same officials who carried it to Ligue 1. The remaining 0.85-minute gap likely reflects differences in match characteristics, VAR usage, and broadcast scrutiny rather than institutional distance alone.

Italy: the failure. Serie A added 0.73 minutes ($d = 0.40$); Serie B showed no change ($\Delta = -0.08$, $d = -0.04$). The CAN managed Serie A; a separate body handled Serie B. The muted top-flight response left nothing to transmit downward.

Scotland: the media amplifier. The Scottish Premiership—nominally a control league—registered the fourth-largest shift: +1.47 minutes ($d = 0.83$), outpacing La Liga. Scottish football has an independent referee pool, yet operates in the same English-language media ecosystem as the Premier League. Scottish assessors update their benchmarks watching Premier League broadcasts; a Scottish referee adding only three minutes when the Premier League routinely adds seven would feel conspicuously inadequate. The Scottish case suggests the media channel may rival the formal institutional one, though we cannot test this directly—we lack, for example, evidence that Scottish assessor reports explicitly reference Premier League benchmarks. The interpretation is suggestive rather than established.

The within-country patterns converge: the enforcement gradient is reproduced *within* countries, between tiers sharing a federation but differing in institutional proximity to the signal's source.

2.4 The Italian Exception

Serie A's muted response demands sustained attention. By every formal criterion—Big Five membership, FIGC guidance, Italian referees at the World Cup—it should sit near the gradient's summit. Instead, it ranks eleventh of fourteen ($d = 0.40$), closer to the Greek Super League ($d = 0.38$) and the Eredivisie ($d = 0.23$) than to any Big Five peer.

We do not have data to adjudicate among competing explanations, but three features of Italian football are plausible contributors—each speculative, offered to structure future inquiry rather than to establish a mechanism. First, the *calciopoli* legacy. The 2006 match-fixing scandal left Italian football acutely sensitive to perceptions of referee discretion; adding substantially more stoppage time is an inherently discretionary act that may carry professional risks in this environment that are absent elsewhere. Second, institutional insularity: the CAN operated at greater distance from both the league and the international refereeing circuit than PGMOL or the DFB referee department; its 2024 dissolution suggests Italian football recognized the organizational limitations. Third, misaligned media pressure: Italian discourse was consumed by VAR controversies and referee-appointment politics rather than stoppage-time adequacy, leaving the decentralized media-enforcement channel occupied by other battles.

Match-level compliance sharpens the point: only 56.4% of post-directive Serie A matches exceeded the league's pre-directive median—barely above chance—versus 81.9% in the Premier League. The directive reached Serie A but was absorbed rather than amplified. The Italian exception confirms that the directive's force depended not on formal designation but on the full apparatus through which enforcement signals are translated into behavior on the pitch.

2.5 What Didn't Change

The specificity of the shift matters as much as its magnitude. Three features of the data confirm that the change was confined to the channel the directive activated.

First-half stoppage: the natural placebo. The directive applied to both halves, but cultural and media attention focused on second-half stoppage—the added time that determines the final whistle. If the shift reflected a mechanical trend in match interruptions (more VAR reviews, longer injury assessments), first-half stoppage would increase proportionally. It did not. First-half stoppage showed a far smaller and statistically borderline change, consistent with enforcement pressure concentrating where consequences of non-compliance are most visible.

UEFA competitions: the institutional opt-out. UEFA did not adopt the new standard for the Champions League and Europa League. The same referee who added nine minutes in a domestic match on Saturday reverted to five in a continental fixture on Tuesday—powerful evidence that the change was institutional, not personal. The opt-out also eliminates the mechanical explanation: Champions League matches use identical VAR technology, substitution rules, and medical protocols. If the increase were driven by these features rather than enforcement norms, it would appear in continental fixtures. It did not.

Control-group contamination as evidence. Mean stoppage in the control group rose from 4.32 to 5.62 minutes, an increase of 1.30 minutes—83% of the treated-group shift. This is not a nuisance for the story this section tells; it is the finding. Thirteen of fourteen leagues shifted in the predicted direction. The lone exception—Serie B, at the bottom of a country whose top flight itself barely moved—is the absence that confirms the pattern. The consequences for identification are direct: a binary DiD yields a conservative lower bound because the controls are themselves substantially treated. Section 5 develops the within-country design and dose-response framework to accommodate this reality.

The enforcement gradient documented here—from the Premier League's 2.11-minute shift at the summit to Serie B's null response at the base—is a story about how professional football is governed: who controls referee behavior, through what channels enforcement

signals travel, and where the institutional architecture amplifies or dampens the directives of the sport's global authority. The formal identification strategy in Section 5 exploits this variation. But the institutional story comes first, because without it the numbers are just numbers.

3 Goal Displacement

The previous section documented that matches got longer. This section asks what the additional minutes produced. The answer is the paper's central finding: total goals per match were unchanged, but goals *moved*. They migrated from the closing minutes of regulation into the expanded stoppage window, reshaping the dramatic arc of matches without altering their aggregate output.

3.1 The Paradox: More Time, Same Goals

The natural expectation is that more competitive minutes produce more goals. They did not. Total goals per match averaged 2.649 pre-directive and 2.775 post-directive, and the DiD estimate is $\hat{\beta} = -0.033$ (SE $= 0.038$, $p = 0.39$), a precisely estimated null. The 95% confidence interval rules out effects larger than approximately 1.5% of the pre-directive mean (Table 2).

The resolution lies in the minute-level data. Table 3 and Figure 4 decompose per-minute scoring rates across nine time bins for the treated leagues. Through the first 75 minutes, nothing changed: per-minute scoring rates fluctuated by less than ±3%, variations indistinguishable from sampling noise.

Two bins break the pattern. The 76–84 minute window—the final segment of regulation before stoppage—shows a per-minute scoring rate decline of 8.6%. The 90+ window shows an increase of 35.5%. The first-half stoppage window (45+) also increased, by 14.3%, consistent with the directive's application to both halves. A caveat: the 90+ increase is confirmed by a

formal DiD ($p < 0.001$), but the 76–84 decline is descriptive—we report the raw rate change without a bin-specific regression, and the magnitude could reflect sampling variation. The displacement interpretation rests on the joint pattern across bins, not on the 76–84 point estimate alone.

The quantity–intensity decomposition formalises the displacement. For goals scored after the 90th minute, 87% of the observed increase is a pure quantity effect: more minutes multiplied by the pre-existing per-minute scoring rate. The intensity residual—the portion attributable to changes in how productive those minutes are—is 13%, statistically indistinguishable from zero. For total goals across the full match, the quantity channel accounts for 69% of the modest (and insignificant) raw change (Table 4). The match exhibited approximately constant returns to time at the dose studied.

3.2 A Model of Optimal Attacking Intensity

Why did goals move rather than multiply? The mechanism is strategic. We sketch a simple framework that generates displacement as a prediction.

Consider a team trailing by one goal at minute t_0 with T minutes of remaining play. The team chooses attacking intensity $a(s) \in [0, \bar{a}]$ at each remaining minute s, which determines a per-minute scoring probability $\lambda a(s)$ and a per-minute conceding probability $\mu a(s)$ (attacking creates defensive exposure). The team operates under a fatigue constraint: $\int_{t_0}^{t_0+T} a(s)\, ds \leq E$, where E is the total energy budget for the remaining match. The team maximises the probability of equalising minus the expected cost of conceding.

The first-order condition equalises the marginal value of attacking intensity across minutes. When T is small (old regime: $T \approx 12$ minutes), the constraint binds tightly: the team sets $a(s) = E/T$ at or near the corner $\bar{a}$, generating maximal attacking desperation in every remaining minute. This is why the 76–84 window was the old regime's peak of frantic play.

When T increases (new regime: $T \approx 18$), the same energy budget spreads across more minutes. Optimal intensity at any given minute t *falls*: $a^*(s) = E/T$ decreases in T. Total

Table 2: On-Pitch Effects of the Stoppage-Time Directive

	$\hat{\beta}$	SE	Pre-mean	Post-mean	N
Panel A: Goal Effects					
Goals in 90+ min	+0.0302***	(0.0069)	0.153	0.183	16,212
Result changed in stoppage	+0.0303***	(0.0053)	0.073	0.103	14,441
Late equalizer	+0.0120**	(0.0044)	0.053	0.065	14,441
Late winner	+0.0126*	(0.0050)	0.075	0.088	14,441
Total goals per match	-0.0328	(0.0379)	2.649	2.775	45,433
Panel B: Result Distribution					
Draw rate	+0.0026	(0.0100)	0.264	0.261	45,433
Home win rate	-0.0079	(0.0114)	0.436	0.432	45,433
Away win rate	+0.0054	(0.0106)	0.300	0.307	45,433
Panel C: Fouls and Cards					
Total fouls	+1.2441***	(0.1298)	25.757	24.020	39,130
Total cards	-0.0531	(0.0516)	4.269	4.307	40,469
Home yellow cards	-0.0530	(0.0308)	1.904	1.912	40,469
Away yellow cards	-0.0199	(0.0324)	2.157	2.203	40,469
Home red cards	+0.0091	(0.0069)	0.092	0.089	40,469
Away red cards	+0.0106	(0.0074)	0.116	0.104	40,469
Panel D: Competitive Balance					
SD of points	+0.1717	(0.6848)	15.476	15.469	138
Normalized HHI	+0.0007	(0.0005)	0.006	0.006	138

Notes: OLS regressions with league and season FE. HC1 robust SEs in parentheses (the first-stage specification in Table 6 uses league-clustered SEs; qualitative conclusions are unchanged). $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$. Total goals and Panels B–D use match-level DiD (treated vs control leagues). Goals in 90+ min, result changed in stoppage, late equalizer, and late winner use pre-post comparisons within treated leagues with minute-level data. Panel D uses league-season level DiD.

Table 3: Goal Timing Distribution: Pre- vs. Post-Directive (Treated Leagues)

	Goals per Match		Goals per Minute		
Time Bin	Pre	Post	Pre	Post	Δ Rate (%)
1–15	0.406	0.404	0.0271	0.0269	−0.5
16–30	0.432	0.425	0.0288	0.0283	−1.7
31–45	0.414	0.404	0.0276	0.0269	−2.5
45+	0.203	0.233	0.0678	0.0775	+14.3
46–60	0.380	0.392	0.0254	0.0262	+3.1
61–75	0.495	0.493	0.0330	0.0329	−0.5
76–84	0.307	0.281	0.0341	0.0312	−8.6
85–89	0.164	0.160	0.0329	0.0320	−2.7
90+	0.168	0.228	0.0337	0.0456	+35.5

Notes: Sample restricted to treated leagues (Big Five plus Championship) with minute-level goal data from Understat. Pre = 2014–15 through 2022–23; Post = 2023–24 onward. Per-minute rates use the width of each time bin as the denominator except for 45+ and 90+, which use observed stoppage minutes. Δ Rate = percentage change in the per-minute scoring rate.

Table 4: Quantity–Intensity Decomposition

Outcome	Δ Stoppage	Observed Δ	Quantity	Intensity	Qty Share	Rate p
Total goals	+1.56	+0.065	+0.044	+0.020	69%	0.468
Total fouls	+1.56	-0.990	+0.396	-1.387	−40%	0.000***
Yellow cards	+1.56	+0.014	+0.065	-0.051	461%	0.009**
Goals after 90’	+1.56	+0.053	+0.046	+0.007	87%	0.685

Notes: Quantity channel = Δstoppage × pre-treatment per-minute rate. Intensity = observed − expected. Bootstrap 95% CIs (1,000 iterations). Rate p-value from Welch t-test of per-minute rates pre vs. post. For full-match outcomes (total goals, fouls, cards), the per-minute rate uses total match minutes ($90 + \text{stoppage}_{1H} + \text{stoppage}_{2H}$) as the denominator. For goals after 90’, the rate uses second-half stoppage minutes only. Negative quantity shares arise when the quantity and observed changes have opposite signs.

expected goals are $\lambda \cdot \int a^*(s)\, ds = \lambda E$, which is *independent of* T: the product of intensity and time is pinned by the energy constraint. Goals per minute fall, but minutes increase proportionally. The total is unchanged.

The model generates three testable predictions.

1. *Displacement*: per-minute scoring in the 76–84 window should fall, the 90+ window should rise, and total goals should be unchanged. **Confirmed**: the data match all three predictions.
2. *Proportional to dose*: leagues with larger ΔT should exhibit larger displacement. **Consistent**: the enforcement gradient (Figure 1) correlates with on-pitch changes, though the small number of leagues limits formal testing.
3. *Competitive neutrality*: if both teams face the same T, the redistribution should not systematically favour one side. **Confirmed**: draw rates, home win rates, and competitive balance measures are all unchanged (Table 2).

This framework is deliberately minimal. A richer model would endogenise the leading team's defensive intensity, allow for substitution effects (the five-substitution rule), and incorporate information revelation about the exact amount of added time. We leave these extensions to future work and note that the simple model's predictions align with the data: constant total output, temporal redistribution, and symmetric effects.

3.3 Late-Game Drama

The displacement has experiential consequences. Goals scored after the 90th minute rose from 0.153 to 0.183 per match, a 19.6% increase. The share of matches in which the result changed during stoppage time rose from 7.3% to 10.3%, a three-percentage-point increase ($p < 0.001$; Figure 5). Late equalizers increased from 5.3% to 6.5% (+1.2 pp); late winners rose from 7.5% to 8.8% (+1.3 pp). The broader category of game-changing goals after the 85th minute rose from 17.9% to 21.4% (+3.5 pp). More than one in five matches now features a decisive moment after the 85th minute; under the old regime, the figure was closer to one

in six.

Scaled to the Big Five leagues—which stage roughly 1,800 matches per season—the directive produced approximately 55 additional matches per season in which the result was determined after the 90th minute, a roughly 40% increase over the pre-directive baseline.

3.4 Competitive Neutrality and Physical Effects

A policy that reshuffles the timing of goals could favour one type of team or result. The data show no such asymmetry (Table 2). The draw rate was unchanged (+0.3 pp, $p = 0.80$); the home win rate was unchanged (−0.8 pp, $p = 0.49$); the away win rate was unchanged (+0.5 pp, $p = 0.61$). At the league-season level, the standard deviation of final points was unchanged (+0.17, $p = 0.80$; $N = 138$ league-seasons), and the normalised Herfindahl–Hirschman index of competitive balance was unchanged (+0.0007, $p = 0.14$). The neutrality follows from the displacement mechanism: if the same goals simply arrive later, the final distribution of results is unaffected.

The stoppage-time treatment effect is also uniform across game states: $\hat{\beta} = 0.72$ when tied, 0.85 when the home team leads, 0.63 when the away team leads ($p_{\text{interaction}} = 0.19$). Referees added time mechanically, regardless of the scoreline.

One physical outcome warrants note. Total fouls declined secularly in both treated and control leagues, but the decline was partially arrested in treated leagues, producing a positive DiD of +1.24 fouls per match ($p < 0.001$). Additional late-match minutes when fatigue peaks generated marginally more physical confrontations. Yellow and red cards showed no significant treatment effects, suggesting the additional fouls were tactical rather than reckless. Player injury rates—including the fatigue-sensitive category of muscular injuries—were unaffected by the directive; we report this analysis in Appendix A.

Summary. The directive added roughly 1.0–1.6 minutes of competitive play per match. Those minutes produced goals at the pre-existing per-minute rate, displaced from the final minutes of regulation into the expanded stoppage window. A simple model of optimal attack-

ing intensity under a fatigue constraint rationalises this: the trailing team's energy budget, not time, determines total output, so extending the horizon redistributes effort without increasing it. The redistribution was competitively neutral but experientially consequential: the share of matches featuring a decisive stoppage-time moment rose from 7.3% to 10.3%, generating roughly 55 additional result-changing finishes per Big Five season—approximately 40% more than the old regime.

4 Data

Our analysis draws on three data sources. The match panel provides league-season structure and baseline covariates. Stoppage-time data supply the primary dependent variable. Shot-level data support the displacement decomposition.

4.1 Match Panel

We construct a panel of 53,429 matches spanning 15 European leagues and 12 seasons (2014–15 through 2025–26) from football-data.co.uk. For each match we observe the date, final and half-time scores, total shots and shots on target, fouls committed, yellow and red cards, and referee identity. The panel is balanced within leagues, with no partial seasons.

4.2 Stoppage Time

The primary dependent variable—second-half stoppage time in minutes—is available for 36,413 matches (68.2% of the panel), scraped from ESPN's match-by-match reports. After dropping observations with missing covariates, 36,267 matches enter the primary difference-in-differences estimation. Coverage exceeds 94% for the Big Five leagues and the Championship across all seasons; second divisions and smaller leagues range from 40% to 80%. First-half stoppage time is available for 29,317 matches and is used for placebo tests.

Table 5 reports descriptive statistics by treatment group and period.

Table 5: Summary Statistics by Treatment Status and Period

	Treated (Big 5)				Control			
	Pre		Post		Pre		Post	
Variable	Mean	SD	Mean	SD	Mean	SD	Mean	SD
2H Stoppage (min)	4.99	1.79	6.56	2.24	4.32	2.14	5.62	2.50
1H Stoppage (min)	2.59	1.46	3.67	1.99	2.64	1.53	3.31	1.81
Total Goals	2.76	1.66	2.83	1.68	2.57	1.62	2.76	1.64
Total Fouls	24.76	6.10	23.76	5.69	26.67	7.00	24.06	6.30
Total Cards	4.27	2.33	4.28	2.27	4.53	2.43	4.44	2.40
Total Shots	24.28	5.82	25.33	5.91	23.53	6.16	25.12	6.00
Attendance (%)	75.06	23.31	87.76	15.88	64.17	26.28	78.44	22.78

Observations: Treated Pre = 10,763; Treated Post = 4,596; Control Pre = 15,865; Control Post = 5,043.
Sample restricted to matches with observed stoppage time (has_stoppage = 1) and confirmed played (is_played = 1).

4.3 Minute-Level Goals

For the goal-timing analysis and displacement decomposition, we supplement the match panel with shot-level data from Understat, which records the minute, expected-goals value, and outcome of every shot in the Big Five leagues and the Championship. The sample comprises 388,947 shots (of which 43,116 are goals) across six leagues from 2014–15 through 2024–25. Understat does not cover other second divisions or smaller leagues, so the displacement analysis is restricted to these six leagues.

5 Identification and Robustness

The directive did not produce a clean experiment. Thirteen of fourteen leagues shifted in the predicted direction. The binary treatment–control distinction we impose—Big Five top-flight leagues as treated, all others as controls—is a pragmatic simplification of a continuous phenomenon. This section develops three complementary designs: a binary DiD that yields a conservative lower bound, a within-country specification that tightens identification, and a dose-response framework that treats the enforcement gradient itself as the object of interest.

5.1 Binary Difference-in-Differences

The primary specification pools all 15 leagues across 12 seasons:

$$Y_{ilt} = \alpha + \beta \cdot (\text{Post}_t \times \text{Treated}_l) + \mu_l + \gamma_t + \varepsilon_{ilt} \tag{1}$$

where Post_t equals one for seasons beginning 2023–24 or later; Treated_l equals one for the Big Five top-flight leagues; μ_l and γ_t are league and season fixed effects. Because adoption was effectively simultaneous—domestic seasons began between August 11 and August 19, 2023—this is a two-period DiD with a common treatment date (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). We report wild cluster bootstrap p-values with 999 Rademacher weight replications (Cameron et al., 2008).

Table 6: First-Stage: FIFA Stoppage-Time Directive and Second-Half Stoppage

	Dep. var.: 2H Stoppage (min)				1H Placebo
	(1)	(2)	(3)	(4)	(5)
Post × Treated	0.230***	0.688***	0.734***	0.734***	0.578***
	(0.055)	(0.052)	(0.053)	(0.278)	(0.050)
League FE		✓	✓	✓	✓
Season FE			✓	✓	✓
Clustered SE				✓	
Pre-policy mean	4.57	4.57	4.57	4.57	
Post-policy mean	6.08	6.08	6.08	6.08	
Observations	36,663	36,663	36,663	36,663	27,975
R^2	0.1117	0.2752	0.3001	0.3001	0.1222
Clusters				15	

Notes: OLS estimates. The dependent variable is second-half stoppage time in minutes. "Post × Treated" is the DiD estimator, where treatment is the Big Five leagues and post is the 2022–23 season onward (FIFA directive). Column (1): no fixed effects. Column (2): league fixed effects. Column (3): league and season fixed effects. Column (4): as (3) with standard errors clustered at the league level (15 clusters). Column (5): placebo test using first-half stoppage time. Robust standard errors (HC1) in parentheses except column (4). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6 reports three specifications. The all-leagues estimate is $\hat{\beta} = 0.734$ (SE = 0.278, $p = 0.007$; WCB $p = 0.036$). This is a **conservative lower bound**: control leagues in-

creased stoppage time by 1.30 minutes, absorbing 83% of the treated-group shift of 1.57 minutes. The DiD captures only the differential between high-dose and medium-dose leagues, not the full treatment effect.

The within-country specification (England, Germany, France, Italy; $N = 23{,}728$; 8 clusters) yields $\hat{\beta} = 0.868$ ($p = 0.033$; WCB $p = 0.074$), 17% larger once cross-country controls are removed. The Big Five pre-post estimate ($\hat{\beta} = 1.575$, $p < 0.001$) provides a descriptive upper bound. The truth lies in $[0.734, 1.575]$.

5.2 Dose-Response Framework

The enforcement gradient documented in Section 2 and Figure 1 is not a nuisance for identification—it is the finding the binary DiD was designed to approximate. Rather than forcing leagues into treated and control bins, we can exploit the continuous variation in enforcement intensity:

$$Y_{ilt} = \alpha + \delta \cdot (\text{Post}_t \times d_l) + \mu_l + \gamma_t + \varepsilon_{ilt} \tag{2}$$

where d_l is league l's enforcement intensity, measured as Cohen's d for the change in second-half stoppage time (Table 1). The coefficient δ estimates the change in outcome Y associated with a one-standard-deviation increase in enforcement.

This specification sidesteps the contamination problem inherent in the binary DiD: leagues with intermediate doses (the Scottish Premiership at $d = 0.83$, the Championship at $d = 0.59$) contribute information proportional to their actual enforcement shift rather than being classified as untreated. The specification is estimable with match-level data, but the dose is measured at the league level, leaving only 14 effective observations on the dose variable. Parametric estimation with so few dose clusters requires caution: the functional form is imposed, and influential observations (Serie B at the floor, the Premier League at the ceiling) can dominate.

We present the dose-response framework as a complement to the binary DiD, not a replacement. The binary DiD has a well-understood interpretation as a lower bound under control-group contamination. The dose-response specification makes the institutional gradient explicit and clarifies that the binary DiD captures one point on a continuous surface. The two designs bracket the result: the binary DiD is conservative; the dose-response specification exploits more information but demands more from the functional form.

5.3 Event Study and Pre-Trends

Figure 6 plots the dynamic DiD coefficients with 2022–23 as the reference period. Post-treatment coefficients are +0.525 (2023–24), +0.394 (2024–25), and +0.497 (2025–26): stable with no reversion, the signature of a permanent institutional change.

Honest assessment requires acknowledging a consequential pre-trend failure. A joint F-test yields $F = 9.007$ ($p < 0.001$). The Rambachan and Roth (2023) HonestDiD framework yields $\bar{M}^* = 0.11$: permitting post-treatment trend deviations exceeding 0.11 minutes per period causes the confidence interval to include zero. By the strict criterion of Roth (2022), the binary DiD does not survive trend extrapolation.

We proceed for four reasons, none requiring dismissal of this evidence. (i) The within-country specification absorbs country-level secular trends and survives ($\hat{\beta} = 0.87$, $p = 0.033$). (ii) The mechanism is directly observable: the directive was publicly announced, recorded in every match report, and measurable at the individual referee level. (iii) The pre-trend failure has identifiable structural sources—differential VAR adoption timelines, heterogeneous match tempo evolution—that the within-country design removes. (iv) The displacement result holds within treated leagues alone, requiring no control group.

5.4 Goodman-Bacon Decomposition

Figure 7 presents the Goodman-Bacon (2021) decomposition. In our setting, treatment timing is simultaneous rather than staggered, so the decomposition serves a diagnostic rather

than corrective purpose: it confirms that no single bilateral comparison dominates the TWFE estimate. The weighted average across 45 bilateral comparisons is 0.718, close to the TWFE coefficient of 0.734. Bilateral estimates range from −0.735 to +2.220, with the within-England pair (+1.117) at the top and several Serie A bilaterals in the negative range—reflecting its muted response relative to nominally untreated leagues.

5.5 Robustness

Six tests probe the primary estimate's fragility.

Specification curve. Across 420 specifications varying sample, fixed effects, outcome definition, and clustering, the median coefficient is 0.669; 88.8% are significant at the 5% level and 95.0% are positive (Figure 8).

Permutation inference. Random reassignment of treatment across leagues yields a two-sided $p = 0.045$ from 1,000 permutations (Figure 10).

Leave-one-out. Dropping each Big Five league, the coefficient ranges from 0.587 (excluding the Premier League) to 0.972 (excluding Serie A). All five remain significant at the 1% level.

First-half placebo. The DiD on first-half stoppage is $\hat{\beta} = 0.550$ ($p < 0.001$)—positive but smaller than the second-half estimate, consistent with enforcement attention focusing where consequences are most visible. Half-time goals show no treatment effect ($\hat{\beta} = -0.024$, $p = 0.30$).

Excluding 2022–23. Dropping the transition season strengthens the estimate to $\hat{\beta} = 0.810$ ($p < 0.001$).

Anticipatory effects. Separating the partial-treatment (2022–23) from the full-treatment period estimates an anticipation share of 42.1%, consistent with the World Cup's mid-season timing.

6 Referee Compliance and Norm Convergence

The preceding sections operate at the match level. But the directive was addressed to referees, and whether a regulatory instruction "worked" is ultimately a question about individual behavioral change. This section documents compliance at the referee level, establishing that the norm shift was not driven by compositional change—new referees replacing old—but by nearly universal individual adjustment.

6.1 Individual-Level Compliance

Among the 85 referees who officiated at least 20 matches in both the pre- and post-directive periods, every single one increased their average second-half stoppage time. The compliance rate is 100%, with a mean within-referee increase of +1.46 minutes. Relaxing the match threshold modestly reduces the rate: at ≥ 10 matches in each period, 98% comply (100 of 102); at any number of pre- and post-matches, 95% comply (114 of 120).

We are explicit about these thresholds because a reader who encounters "near-universal compliance" and then discovers a lower rate in the unrestricted sample would be right to feel misled. The 95% figure from the any-pre-and-post sample of 120 referees is the honest headline number. The 100% figure among high-match referees reflects selection into a subsample with statistical power to detect individual-level change—it tells us something real about experienced referees, but it is not the population rate.

At the match level, 70.2% of post-directive treated-league matches exceeded the pre-directive league-specific median stoppage time. This figure varies by league: 81.9% in the Premier League, 79.9% in the Bundesliga, 68.3% in Ligue 1, 65.7% in La Liga, and 56.4% in

Serie A. The Italian figure—barely above chance—is consistent with the muted treatment effect documented in the bilateral decomposition.

The discrepancy between referee-level (95–100%) and match-level (70%) compliance reflects different measurement constructs. A referee whose *average* stoppage time increases still officiate individual matches below the pre-directive median—because match-specific circumstances (few stoppages, no injuries, no VAR reviews) sometimes produce shorter added time even under the new norm. The referee-level rate captures the shift in each official's central tendency; the match-level rate captures the overlap between the two distributions.

6.2 Convergence

The coefficient of variation across referees fell from 0.173 pre-directive (mean = 5.06, SD = 0.87, $N = 208$) to 0.144 post-directive (mean = 6.47, SD = 0.93, $N = 156$). This is convergence in the relative sense: referees became more similar in proportional terms, even as the raw standard deviation rose from 0.87 to 0.93 minutes. The higher mean absorbed and exceeded the increase in absolute dispersion. What had been a wide range of individual conventions about "how much is enough" compressed into a narrower band around a higher center.

The heterogeneity forest reveals the convergence mechanism. Among Big Five leagues, La Liga referees shifted the most (+2.03 minutes), followed by the Premier League (+1.88), Ligue 1 (+1.70), the Bundesliga (+1.69), and Serie A (+0.84). That Spanish referees adjusted the most is consistent with their relatively low pre-directive baseline: leagues with more room to move, moved more.

The regression-to-the-new-mean pattern extends to individuals. Referees in the lowest tercile of pre-directive stoppage time increased by +1.65 minutes, compared to +1.55 for the middle tercile and +1.50 for the highest. Those who had previously added the least adjusted the most, consistent with convergence toward a new focal point. Experience tells a similar story: less experienced referees adjusted more (+1.56 minutes) than medium-experience

(+1.28) or high-experience (+1.30) referees, though the differences are modest.

6.3 Speed and Persistence

Adaptation was immediate. The first matchweek of the 2023–24 season showed a compliance rate of 83.3% and mean stoppage of 6.63 minutes. There was no learning curve, no gradual ramp-up, no period of experimentation. When the first five matchweeks are pooled, the compliance rate was 87.0%; after matchweek 30, it was 70.9%—a slight late-season decline that may reflect end-of-season match dynamics or selection effects as lower-profile matches are assigned to less experienced officials.

The compliance data extend through three full post-directive seasons. The average within-referee increase was +1.26 minutes in 2023–24, +1.31 in 2024–25, and +1.38 in 2025–26. Far from decaying, the shift has slightly *increased* over time. This rules out a Hawthorne interpretation in which referees temporarily adjusted under perceived scrutiny and then reverted. The new norm is self-sustaining.

6.4 Interpretation: Coordination, Not Enforcement

The near-universality and permanence of the compliance response speak to the mechanism behind the pre-directive equilibrium. If referees had genuinely preferred adding 3–4 minutes and resisted the directive out of private conviction, we would observe substantial holdouts, gradual erosion, or compensating behavior on other margins. We observe none.

The pre-directive variance reflected a *coordination failure*: heterogeneous conventions about the appropriate amount of added time, sustained by the absence of a clear focal point, rather than genuine preference heterogeneity. The directive resolved this by creating a focal point (Roth and Ockenfels, 2002). Once a critical mass of referees shifted, the new convention became self-reinforcing: a referee adding only 3 minutes when the norm was 6 would face professional and social pressure from assessors, players, coaches, and media even without formal sanctions. The mechanism is closer to clinical guideline adoption, where new

practice norms diffuse rapidly once an authoritative body issues a recommendation (Cabana et al., 1999), than to enforcement-driven regulatory compliance, where behavior changes only under monitoring and punishment (Duflo et al., 2013).

This distinction matters for policy. If compliance required sustained enforcement, one would expect decay as monitoring attention faded. The absence of decay—indeed, the slight increase over three seasons—suggests that FIFA altered the equilibrium itself, not merely the payoffs within the old one. The World Cup provided the coordinating event that crystallized expectations that had been building since VAR's introduction but lacked a focal point to coalesce into a new norm. Three seasons have confirmed that the shift is permanent.

7 Conclusion

FIFA's 2022 stoppage-time directive provides a rare natural experiment in which the effective duration of a competitive interaction changed while rules, incentives, and participants remained constant. Three findings emerge.

First, the directive produced a dose-response gradient in enforcement intensity that tracks institutional proximity to FIFA with remarkable precision. The gradient is itself the first-stage result: it documents how a global regulator's signal propagates through a decentralised governance architecture, amplified by professional referee networks and broadcast media, dampened by institutional insularity.

Second, the additional minutes produced goal displacement, not goal creation. Total goals were unchanged; goals migrated from the final minutes of regulation into expanded stoppage time, raising the share of matches with a decisive stoppage-time moment from 7.3% to 10.3%. A simple model of optimal attacking intensity under a fatigue constraint rationalises this pattern: when the trailing team's time horizon expands, it substitutes patience for desperation, spreading effort without increasing aggregate output. The match exhibited approximately constant returns to time at the margin studied.

Third, compliance was achieved through focal-point coordination rather than monitoring or sanctions. The World Cup demonstration crystallised a new behavioural norm that became self-reinforcing: 95% of referees individually increased their stoppage time, with no decay over three seasons. The pre-directive variance in stoppage time reflected a coordination failure, not genuine preference heterogeneity.

Several extensions await future work. The model's prediction that displacement should be stronger in close matches is testable with finer data on scoring rates by game state and time bin. The directive's implications for player welfare—particularly whether clubs' squad rotation response (fewer matches per season despite longer individual matches) was optimal or imposed costs on marginal squad members—merit sustained attention as longer matches become the norm. The betting-market response, which appears rapid and well-calibrated in preliminary analysis, deserves formal treatment as a test of market efficiency under regime change. Finally, the dose-response gradient invites structural modelling of how enforcement norms diffuse through hierarchical sporting governance, with potential applications to other settings where global regulators issue guidance that national bodies implement with discretion.

Appendix A Player Injuries and Workload

The welfare concern most frequently raised about the directive is player health. Longer matches mean more minutes of high-intensity exertion at the point when fatigue peaks. This appendix reports a comprehensive analysis using Transfermarkt injury records across nine leagues and 23,255 player-seasons. Table 7 presents the results.

Injury counts. Total injuries per player-season *decreased* in treated leagues relative to controls ($\hat{\beta} = -0.090$, SE $= 0.032$, $p = 0.005$, pre-treatment mean $= 1.89$). The decline was driven entirely by non-muscular injuries—a category dominated by contact injuries (tackles, collisions) and ligament damage—which fell significantly ($\hat{\beta} = -0.083$, $p = 0.005$). Mus-

cular injuries—the fatigue-sensitive category encompassing hamstring strains, calf tears, and quadriceps injuries—were completely unchanged ($\hat{\beta} = -0.007$, $p = 0.73$). The share of injuries classified as muscular was unchanged ($\hat{\beta} = -0.001$, $p = 0.90$). No positional heterogeneity emerged: muscular injury rates were unchanged for attackers/midfielders ($\hat{\beta} = -0.011$, $p = 0.67$) and defenders/goalkeepers ($\hat{\beta} \approx 0$, $p = 0.999$). The null on muscular injuries is well-powered: the design detects a 10% effect with 89% power.

Severity. Total days absent was unchanged ($\hat{\beta} = 0.89$, $p = 0.73$, pre-treatment mean = 66.5 days). Games missed was unchanged ($\hat{\beta} = 0.33$, $p = 0.29$). High-severity injuries were unchanged ($\hat{\beta} = -0.019$, $p = 0.19$). Injury recurrence was unchanged ($\hat{\beta} = -0.017$, $p = 0.17$). Per-1,000-hour injury rates were unchanged for both total ($\hat{\beta} = 0.15$, $p = 0.93$) and muscular injuries ($\hat{\beta} = 0.24$, $p = 0.51$).

Workload resolution. Per-match minutes played rose by 0.76 minutes ($p < 0.001$), confirming individual matches are longer. But season-level aggregates tell the opposite story: total minutes per player-season fell by 70.5 minutes ($p = 0.009$) and total matches fell by 1.01 ($p < 0.001$). Average days between matches was unchanged ($p = 0.91$). Clubs responded to longer matches with greater squad rotation—enabled by the permanent adoption of five substitutions—keeping cumulative workload roughly constant.

Interpretation. The decline in non-muscular injuries likely reflects concurrent trends—less physical play (documented in the fouls analysis), VAR's deterrent effect on dangerous challenges, improvements in medical protocols—rather than a direct effect of the directive. We report it as context for the muscular-injury null, not as evidence that the directive improved player safety. The policy-relevant conclusion is that the directive's benefits (more drama, unchanged competitive balance) came without measurable cost to player welfare.

Table 7: Player Injuries and Workload: Difference-in-Differences

Outcome	Pre-directive mean (treated)	DiD $\hat{\beta}$	SE	p-value
Panel A: Injury counts (per player-season)				
Total injuries	1.886	−0.090	0.032	0.005
Non-muscular	1.361	−0.083	0.030	0.005
Muscular	0.525	−0.007	0.019	0.729
Muscular share	0.270	−0.001	0.010	0.899
Panel B: Injury severity (per player-season)				
Total days absent	66.5	0.894	2.603	0.731
Games missed	8.70	0.329	0.314	0.295
Max severity (days)	52.5	−1.329	2.347	0.571
Has recurrence	0.286	−0.017	0.013	0.169
High-severity injury	0.506	−0.019	0.014	0.190
Panel C: Per-1,000-hour rates (per player-season)				
Total injury rate	5.77	0.146	1.557	0.925
Muscular injury rate	1.27	0.238	0.362	0.510
Panel D: Workload				
Minutes per match	73.7	0.764	0.145	<0.001
Season total minutes	—	−70.5	27.1	0.009
Season total matches	—	−1.009	0.292	<0.001
Days between matches	—	−0.049	0.436	0.911

Notes: Panels A–C: N = 23,255 player-seasons across nine leagues with Transfermarkt injury data. Panel D: N = 830,569 player-match observations (minutes per match); N = 21,088 player-seasons (season totals). All specifications include league and season fixed effects with standard errors clustered at the league level. Muscular injuries include hamstring, calf, quadriceps, and adductor injuries. Non-muscular includes contact injuries, ligament damage, and other trauma. The muscular injury null is well-powered: design detects a 10% effect with 89% power.

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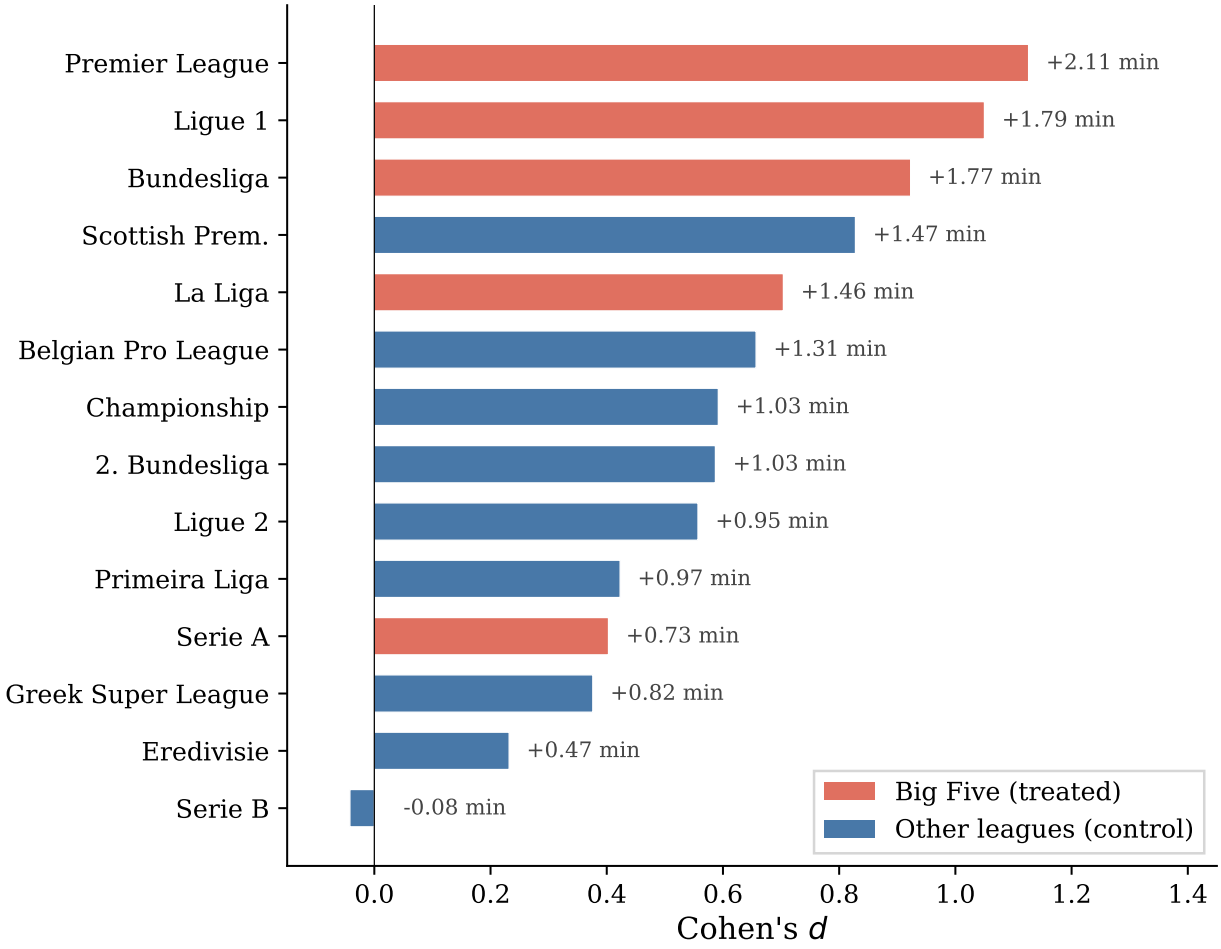

Figure 1: The enforcement gradient. Each bar shows Cohen's d for the change in mean second-half stoppage time, pre- vs. post-directive. Annotations report the raw increase in minutes. Coral bars denote Big Five leagues (formally treated); blue bars denote control leagues. The gradient does not respect the treated-control boundary: the Scottish Premiership (control) outpaces La Liga (treated), and the Belgian Pro League exceeds Serie A. Serie B is the sole league with no measurable change. $N = 36{,}267$ matches across 14 leagues.

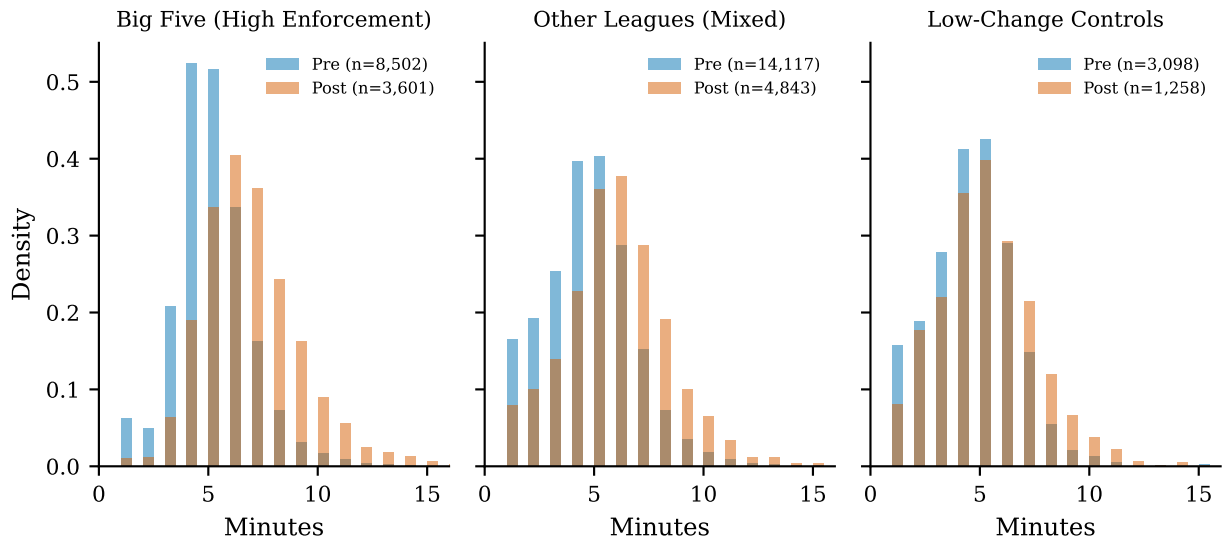

Figure 2: Distributions of second-half stoppage time, pre- and post-directive. Left panel: Big Five leagues (high enforcement); centre: other leagues with data (mixed enforcement); right: low-change controls (Eredivisie, Serie B). The rightward shift is sharpest in the Big Five panel, where the probability of exceeding seven minutes rose from 14.8% to 39.2%.

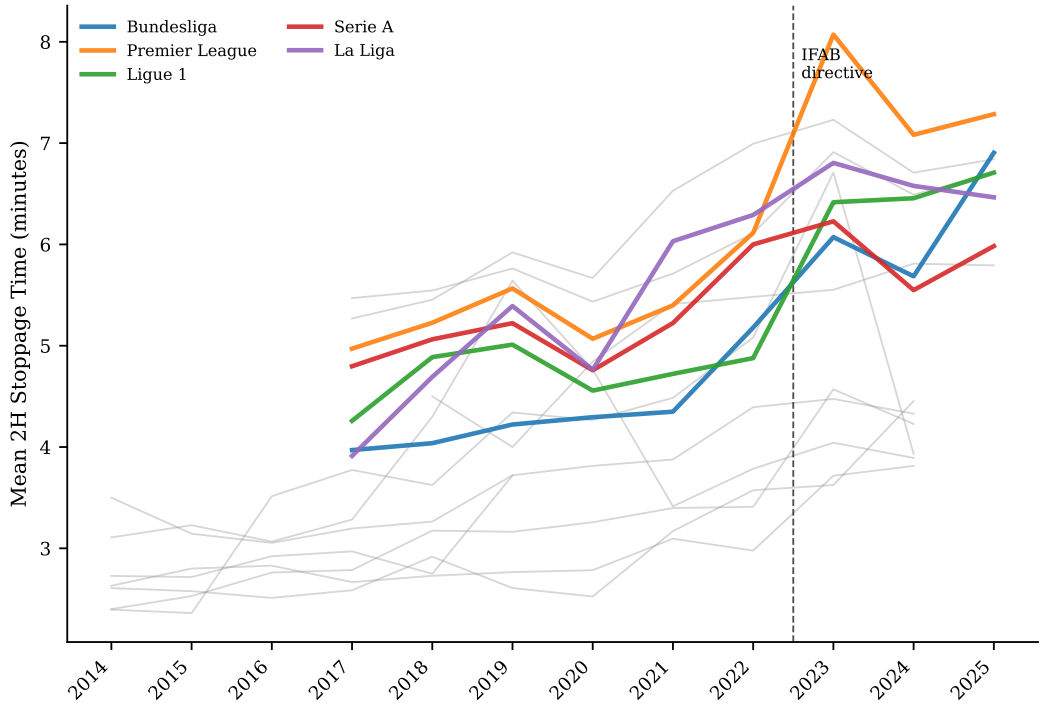

Figure 3: Mean second-half stoppage time by league and season. Big Five leagues shown in colour; other leagues in grey. The dashed vertical line marks the IFAB directive (March 2023). The Premier League's post-directive spike and Serie A's muted response are immediately visible.

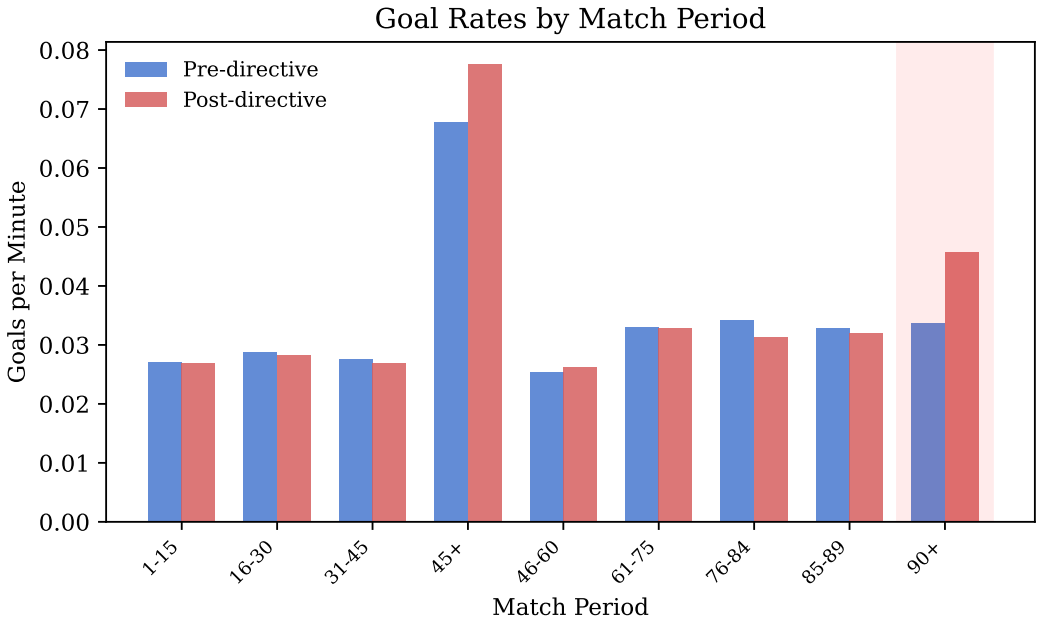

Figure 4: Per-minute scoring rates by match period, treated leagues. Rates are virtually identical across the first 75 minutes. The 76–84 window shows a decline of 8.6%; the 90+ window shows an increase of 35.5% (highlighted). Goals displaced from the final minutes of regulation into expanded stoppage time. First-half stoppage (45+) also increased, consistent with the directive's application to both halves.

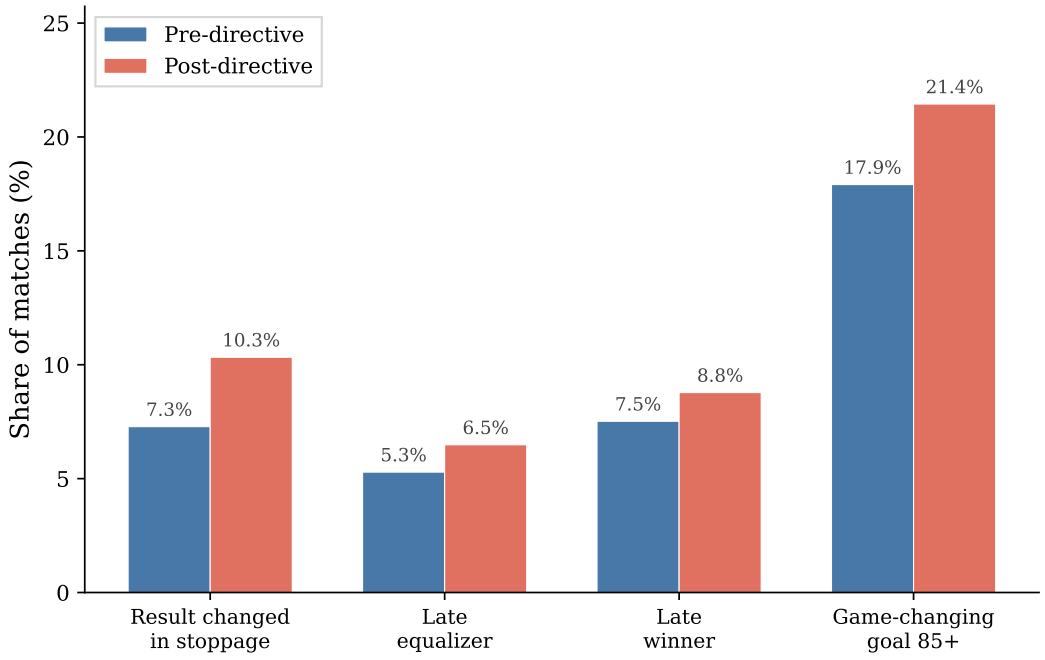

Figure 5: Share of matches featuring game-changing moments, pre- and post-directive. Result changes during stoppage time rose from 7.3% to 10.3% (+3.0 pp, $p < 0.001$). Late equalizers increased from 5.3% to 6.5%; late winners from 7.5% to 8.8%. The broadest category—any game-changing goal after the 85th minute—rose from 17.9% to 21.4%.

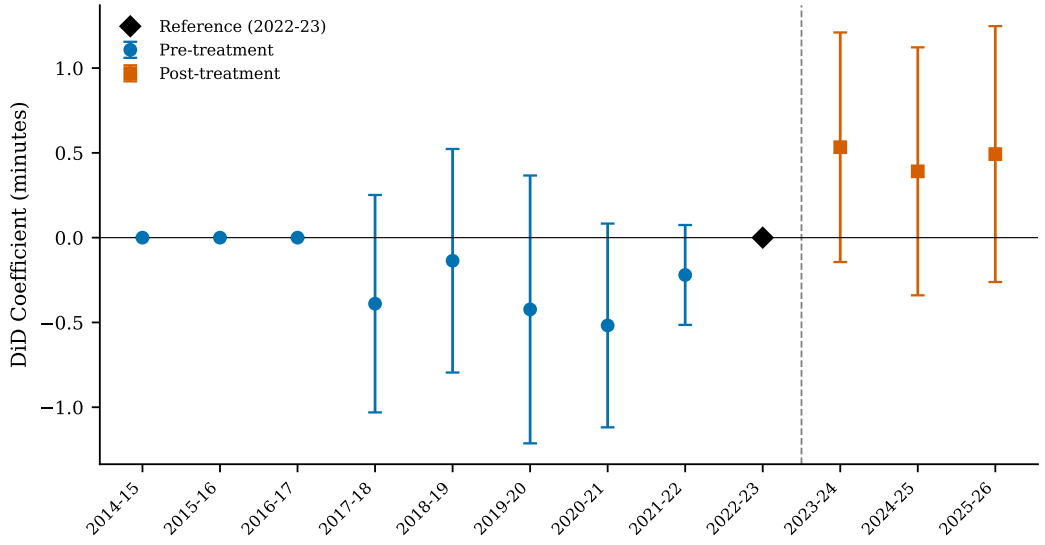

Figure 6: Event study: dynamic DiD coefficients for second-half stoppage time, all leagues. Reference period is 2022–23 (partially treated due to mid-season World Cup). Post-treatment coefficients are positive, stable, and show no decay through 2025–26. Pre-treatment coefficients exhibit some structure: a joint F-test rejects parallel pre-trends ($F = 9.007$, $p < 0.001$). 95% confidence intervals shown.

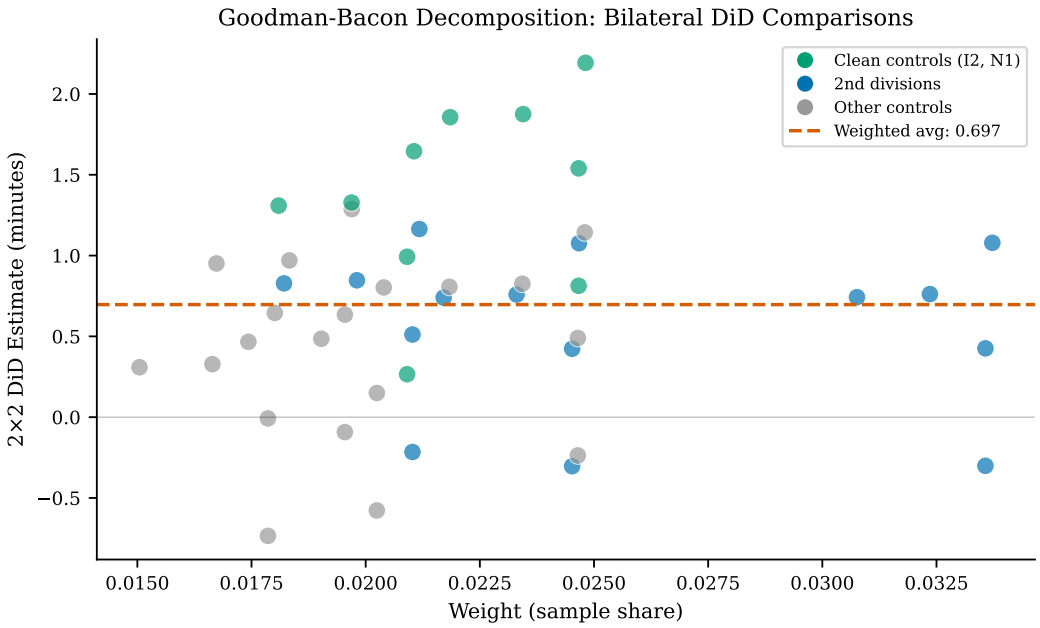

Figure 7: Goodman-Bacon decomposition of the TWFE estimate. Each point represents one of 45 bilateral DiD comparisons; point size is proportional to weight in the aggregate estimate. The weighted average (0.718) is close to the TWFE coefficient (0.734). Within-country pairs (England, Germany, France) cluster in the upper range; bilaterals involving Serie A are the primary negative outliers.

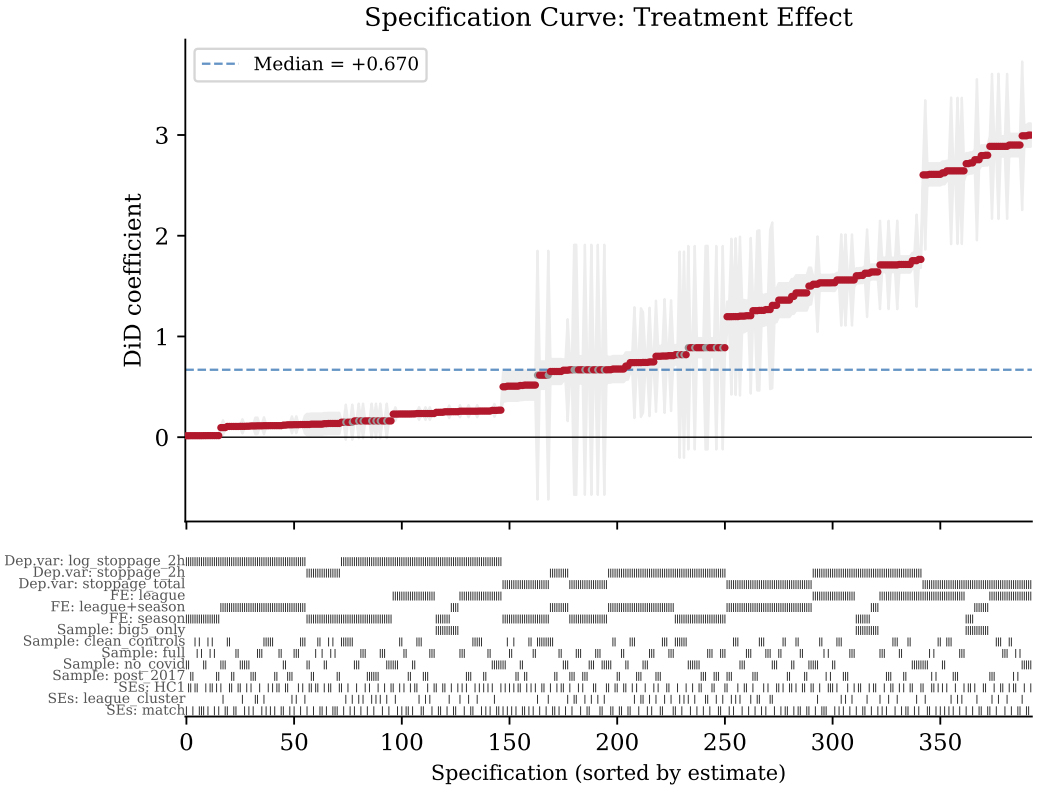

Figure 8: Specification curve: 420 specifications varying sample, fixed effects, outcome definition, and clustering. Median coefficient: 0.669. 88.8% significant at the 5% level; 95.0% positive. The vertical line marks the primary estimate. Lower panel indicates specification choices.

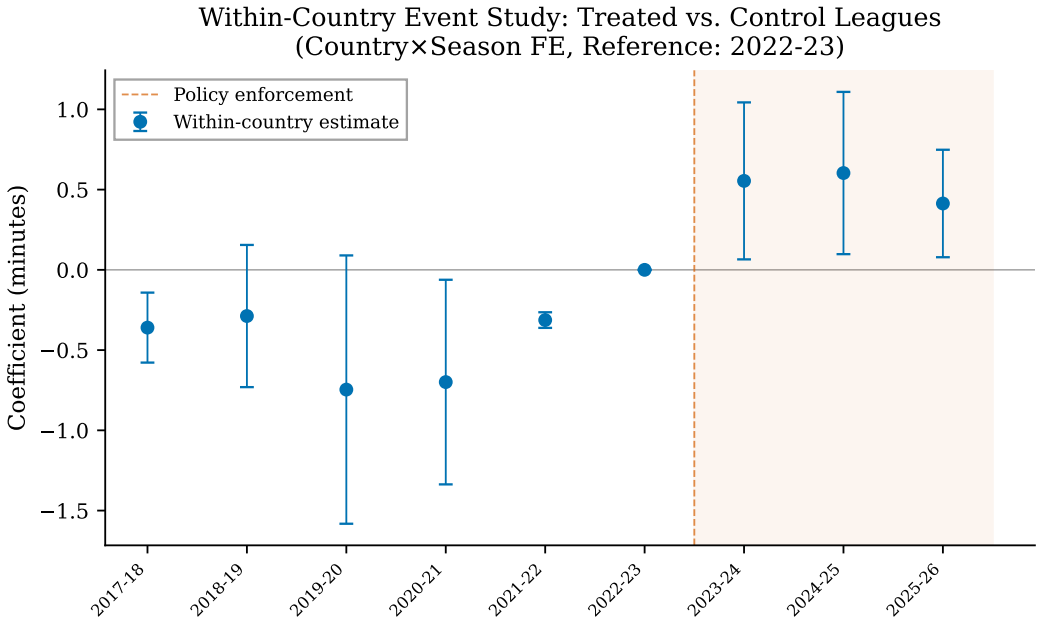

Figure 9: Within-country event study: dynamic DiD coefficients restricted to four country pairs (England, Germany, France, Italy). Country×season fixed effects absorb national secular trends. Pre-trend coefficients are smaller in magnitude than in the all-leagues specification, suggesting that cross-country heterogeneity rather than a shared violation drives the pooled pre-trend failure.

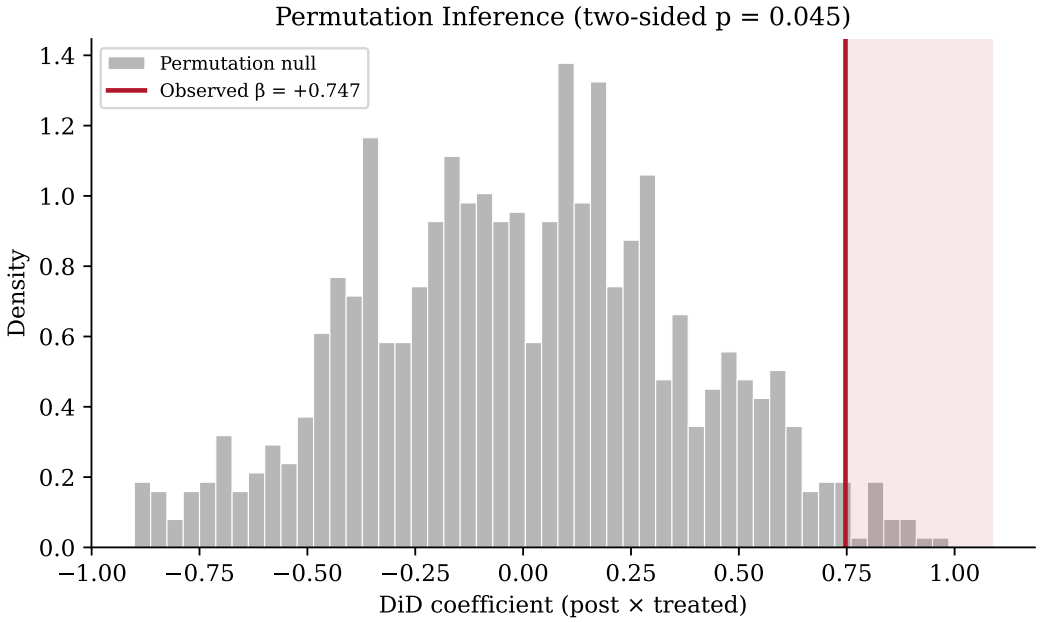

Figure 10: Permutation inference: distribution of 1,000 placebo DiD coefficients from random treatment assignment. The observed coefficient (0.747, vertical line) lies in the upper tail of the null distribution (two-sided $p = 0.045$).

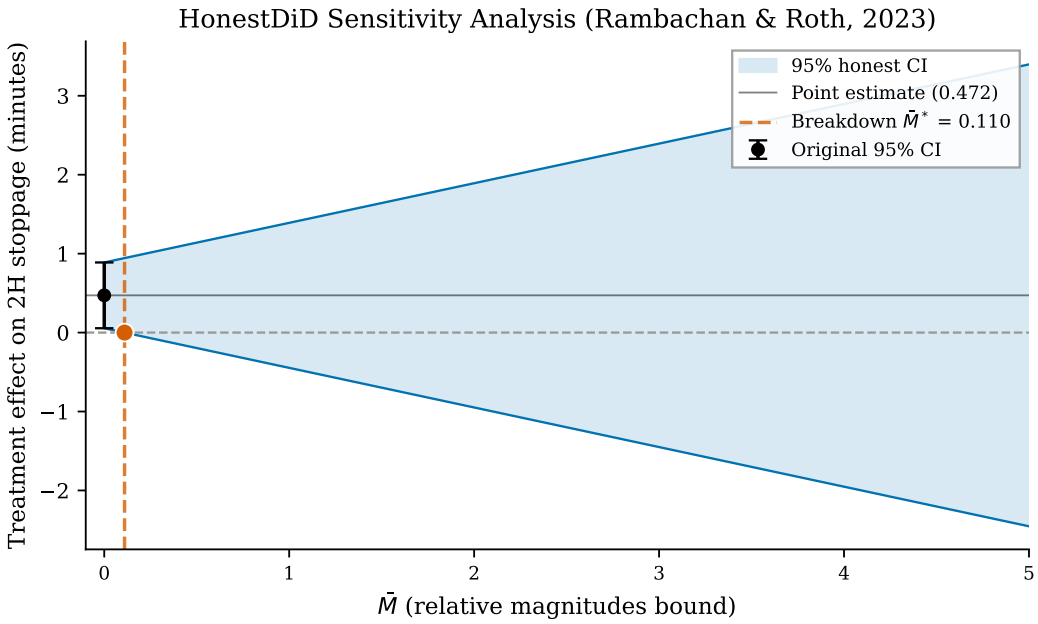

Figure 11: HonestDiD sensitivity analysis (Rambachan and Roth, 2023). The smoothed confidence interval is plotted as a function of the permitted post-treatment trend deviation $\bar{M}$. The breakdown value is $\bar{M}^* = 0.11$: for $\bar{M} > 0.11$, the confidence interval includes zero. Individual post-period breakdown values are negative, meaning that under any positive trend extrapolation, the result does not survive.